
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Out-of-distribution (OOD) detection is a prerequisite for the robustness and reliability of machine learning systems under deployment, where the test-time input distribution is not guaranteed to match the training distribution. A recent line of work exploits unlabeled data to strengthen OOD detection, but the utilization of unlabeled in-the-wild data is impeded by its composition: each wild sample is drawn from a mixture of the in-distribution (InD) and OOD marginals with unknown component membership. Because no clean, separately labeled set of OOD instances is available, estimating an optimal OOD classifier from such a mixture is ill-posed under standard supervised objectives. In this work, we introduce Medix, a framework that extracts candidate outliers from unlabeled data via a coordinate-wise median operation applied to per-sample loss gradients. The median is adopted as the estimator of central tendency because its bounded influence function confers robustness to gradient noise and to a bounded fraction of contaminating outliers, properties we exploit as an OOD-detection mechanism. The extracted outliers, together with labeled InD data, are then used to fit a binary OOD classifier. On the theoretical side, we derive finite-sample error bounds that certify a low misclassification rate for the filtering stage of Medix. The empirical evaluation corroborates these guarantees: Medix consistently outperforms prior methods across open-world detection settings, in agreement with the theoretical analysis.

21 1 Introduction

22 When machine learning models are deployed, the test-time input distribution routinely departs from the training distribution, and this distributional shift manifests concretely as out-of-distribution (OOD) inputs that jeopardize safety and reliability. Such inputs originate from semantic categories absent from the label space $\mathcal{Y}$ and therefore lie outside the region on which the classifier’s predictions are calibrated. Standard neural network classifiers, however, are trained under a closed-world assumption and are equipped with no explicit rejection mechanism, so they emit confident predictions on these inputs rather than abstaining (Nguyen et al., 2015).

29 The core difficulty is that OOD detection must be performed at inference time on distributions to which the model was never exposed during training; consequently, the decision function has no direct supervisory signal for separating OOD from in-distribution (InD) inputs. To supply such a signal, recent approaches (Katz-Samuels et al., 2022a; Du et al., 2024a) augment training with auxiliary “in-the-wild” data. In particular, Katz-Samuels et al. (2022a) regularize model training with unlabeled wild data while retaining the labeled InD classification objective. The appeal of this data source is operational: wild data accrues automatically once a model is deployed in its target environment, and thus provides a low-cost empirical proxy for the test-time OOD distribution, yielding more robust detection.

Exploiting wild data is nonetheless nontrivial, precisely because it is an unlabeled mixture of InD and OOD samples with latent, per-sample component membership. The absence of a clean, separately labeled OOD set is the crux of the difficulty: the detector observes only draws from the marginal mixture and receives no indicator of whether a given sample was generated by the InD or the OOD component. This regime remains comparatively underexplored, and, more critically, is seldom accompanied by formal analysis. To the best of our knowledge, Du et al. (2024a) is the only prior work that supplies a theoretical foundation for the unlabeled “in-the-wild” setting.

In parallel, median-based estimators have proven effective for data pruning (Acharya et al., 2024), owing to their robustness to a bounded fraction of corrupted points. Motivated by these robustness properties, this paper investigates the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

To answer it affirmatively, we develop a median-centric formulation of OOD detection. Concretely, we cast outlier extraction as an optimization problem over subsets of the wild set, and instantiate its solution as an algorithm, Medix, that identifies OOD samples from wild data with a provably low error rate. The algorithm is accompanied by a finite-sample theoretical analysis that certifies this error, and the analysis is subsequently validated empirically; jointly these establish the robustness and computational efficiency of the method. Ours is among the few analyses to furnish such guarantees for the unlabeled “in-the-wild” setting. Specifically, we bound the fraction of InD samples incorrectly flagged as outliers and show that this fraction decomposes into two controllable terms. The *contamination term* quantifies the perturbation induced on the coordinate-wise median by OOD gradients and remains bounded whenever the OOD proportion π satisfies $\pi < 0.5$. The *concentration term* exploits the sub-Gaussian tails of InD gradient coordinates to bound the deviation of individual InD gradients from their mean and decays with sample size. Together, these terms guarantee that, with high probability, the number of misclassified InD points is small, and the bound holds uniformly, including in the worst case.

We evaluate against two method families: (1) detectors trained solely on InD data, and (2) detectors trained on InD data jointly with an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves over the strong KNN+ baseline (Sun et al., 2022) by an average of 40.98 percentage points in FPR95. In contrast to Outlier Exposure (Hendrycks et al., 2019), whose validity rests on the strong distributional assumption that the auxiliary set is fully separable from $\mathbb{P}_{\text{in}}$, Medix attains superior detection without imposing separability, and is therefore applicable under weaker assumptions on the wild data. Relative to WOODS (Katz-Samuels et al., 2022a), Medix reduces the average FPR95 by 1.32 percentage points on CIFAR-100 and 2.60 percentage points on CIFAR-10.

Our key contributions are as follows:

- C1)** We propose Medix, a two-stage procedure that first filters candidate outliers from the unlabeled wild data by greedy minimization of a median-based objective, and then fits a binary OOD detector on the extracted outliers together with the labeled InD samples. The filtering stage is the principal methodological contribution.
- C2)** We establish two-sided theoretical guarantees for median-based filtering, bounding both the rate at which InD samples are flagged as outliers and the rate at which OOD samples are retained as inliers. Specifically, we prove that each misclassification rate is controlled by a contamination term, which stays bounded provided the OOD proportion satisfies $\pi < 0.5$, and a concentration term, which follows from the statistical stability of the InD gradients.
- C3)** We evaluate Medix across eleven InD–OOD pairs against 20 competitive baselines. Medix attains the best performance on every pair. We further verify empirically that the outlier-extraction stage achieves a low error rate (e.g., 12.5%; see Figure 2), in quantitative agreement with the theory.

2 Preliminaries and Problem Setup

This section fixes notation and states the problem formally. We successively specify the labeled InD data-generating process, the OOD-detection objective, the wild-data mixture model, the classifier and loss families, and the evaluation criteria.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{1, \dots, K\}$ the finite label space, so that $K = |\mathcal{Y}|$. The labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ consists of n pairs drawn independently and identically distributed (i.i.d.) from an unknown joint distribution $\mathbb{P}_{XY}$ over $\mathcal{X} \times \mathcal{Y}$. We write $\mathbb{P}_{\text{in}}$ for the marginal of $\mathbb{P}_{XY}$ on $\mathcal{X}$, i.e., the InD input distribution, and use $\mathcal{S}_{\text{in}}$ to fit the InD classifier.

Out-of-distribution detection. We consider the deployment scenario in which a model trained on labeled InD data is subsequently exposed to inputs from semantic classes outside the training label space, i.e., inputs whose true label satisfies $y \notin \mathcal{Y}$; on such inputs the model is required to abstain. The detection objective is therefore a binary decision at inference time: assign each test input either to the InD marginal $\mathbb{P}_{\text{in}}$ or to an OOD source.

Unlabeled wild data. A central obstacle in OOD detection is the scarcity of labeled OOD samples: the support of the OOD distribution is effectively unbounded, rendering the exhaustive collection of labeled examples costly and impractical. Following Katz-Samuels et al. (2022a), we instead incorporate unlabeled wild data $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, which more faithfully reflects the deployment distribution. The wild set is an unlabeled mixture of InD and OOD samples and is collected at deployment time from a model pre-trained on $\mathcal{S}_{\text{in}}$. Following Du et al. (2024a); Katz-Samuels et al. (2022a), we model the marginal distribution of the wild data by the Huber contamination model (Huber, 1964),

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π is the contamination proportion (the mixing weight of the OOD component) and $\mathbb{P}_{\text{out}}$ is the OOD marginal over $\mathcal{X}$. The boundary case $\pi = 0$ places all mass on $\mathbb{P}_{\text{in}}$, so no OOD samples are present and the detection problem is degenerate.

Models and Loss Functions. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier with parameters $\phi \in \Phi$, whose output, after normalization, is a categorical distribution over the $K = |\mathcal{Y}|$ InD classes. The associated multi-class loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For detection, we introduce a distinct scalar-valued classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameters $\theta \in \Theta$, and a binary loss $\ell_b(g_\theta(x), y_b)$ defined over $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. The two binary targets are sign-encoded: $y_+ > 0$ denotes the InD class and $y_- < 0$ the OOD class, so that the sign of $g_\theta(x)$ induces the detection decision.

Learning objective. The learning framework fits the detector g_θ and the multi-class classifier f_ϕ jointly, using both the InD distribution $\mathbb{P}_{\text{in}}$ and the wild mixture $\mathbb{P}_{\text{wild}}$. At test time, performance is quantified by the following three metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ is the OOD test distribution. FPR is to be minimized and TPR and Acc maximized, as indicated by the accompanying arrows.

3 Method: Median-Centric Framework for OOD Detection

This section specifies **Medix**, a learning procedure for OOD detection that exploits unlabeled wild data. Unlike detectors estimated from InD data alone, Medix conditions on the wild mixture and is thus tailored to open-world deployment, where inputs from unseen classes occur at test time. The procedure factorizes into two sequential stages: **(1) Outlier Extraction** — a filtering operator that maps the unlabeled wild set to a subset of candidate OOD samples (Section 3.1); and **(2) Detector Training** — estimation of a binary OOD detector from the labeled InD data together with the candidate outliers returned by stage (1) (Section 3.2), following the training protocol of Du et al. (2024a). As shown in the following subsections, this decomposition both isolates OOD samples from the unlabeled wild data effectively and yields a detector suitable for dynamic, open-world deployment.

3.1 Extracting Candidate Outliers from the Wild Data

To extract candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix operates in the space of per-sample loss gradients rather than in input or feature space. The gradients are computed with respect to the

parameters of a classifier f_ϕ fitted exclusively on the InD dataset $\mathcal{S}_{\text{in}}$. We now specify the construction in three steps: estimation of a reference gradient, the empirical motivation for a median-based criterion, and the resulting subset-selection optimization problem.

Reference gradient estimation from InD data. The first step estimates a reference gradient from the InD dataset $\mathcal{S}_{\text{in}}$. We fit f_ϕ on $\mathcal{S}_{\text{in}}$ by empirical risk minimization (ERM),

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the resulting minimizer. Given the fitted classifier, we define the reference gradient $\bar{\nabla}_{\text{in}}$ as the empirical mean over $\mathcal{S}_{\text{in}}$ of the loss gradient with respect to the model parameters,

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

The vector $\bar{\nabla}_{\text{in}}$ thus serves as a fixed reference against which the gradient of any other sample can be compared, and deviations from it constitute the signal exploited for filtering.

Motivation. The method rests on the hypothesis that increasing the OOD count in the wild set $\mathcal{S}_{\text{wild}}$ monotonically increases the deviation of the wild-set gradient median from the reference gradient $\bar{\nabla}_{\text{in}}$. To test this hypothesis empirically, we conduct a controlled experiment with CIFAR-10 (Krizhevsky et al., 2009) as InD and SVHN (Netzer et al., 2011) as OOD. We initialize $\mathcal{S}_{\text{wild}}$ with 10,000 CIFAR-10 samples held disjoint from the data used to fit $\phi_{\mathcal{S}_{\text{in}}}$ and to compute $\bar{\nabla}_{\text{in}}$, then inject SVHN OOD samples incrementally while recording the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild-set gradients,

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

As reported in Figure 1, the deviation increases monotonically with the number of injected OOD samples, confirming the hypothesis. This monotonicity is not merely diagnostic: it directly motivates both the objective introduced below and the algorithm’s stopping rule, which halts once the change in the L_2 -norm deviation between consecutive iterations falls below a threshold ϵ (detailed in the following subsection).

Filtering potential outliers from unlabeled wild data. Guided by Figure 1, we formulate outlier extraction as the following subset-selection problem, whose solution $\mathcal{S}_{\text{out}}^*$ is the estimated outlier set within $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ is the element-wise (coordinate-wise) median operator, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ is the label predicted by $f_{\phi_{\mathcal{S}_{\text{in}}}}$ for wild sample $\tilde{\mathbf{x}}_i$, used as a pseudo-label since wild samples are unlabeled. For brevity we write $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$ for the set of gradients indexed by $\mathcal{S}$. The objective in (4) selects the subset $\mathcal{S}$ of $\mathcal{S}_{\text{wild}}$ whose gradient EWM is closest, in L_2 distance, to the reference gradient $\bar{\nabla}_{\text{in}}$. By the monotonicity established in Figure 1, the minimizer $\mathcal{S}_{\text{in}}^*$ is a consistent estimator of the InD component of $\mathcal{S}_{\text{wild}}$, and its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ recovers the OOD component.

Exact minimization of (4) is combinatorial and scales prohibitively with $|\mathcal{S}_{\text{wild}}|$. We therefore approximate the minimizer by a greedy, leave-one-out procedure, specified in Algorithm 1, which iteratively removes the samples whose deletion most reduces the gradient-median deviation from $\bar{\nabla}_{\text{in}}$.

At each iteration, the algorithm evaluates the EWM of the gradients of the samples currently retained in the wild set $\mathcal{S}$ and measures its L_2 distance to $\bar{\nabla}_{\text{in}}$, denoted $d_t := \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. For each

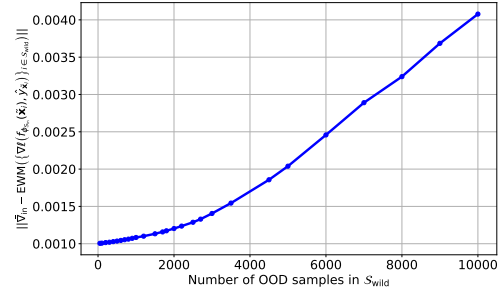


Figure 1: L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the EWM of the wild-set gradients as the number of injected OOD samples in $\mathcal{S}_{\text{wild}}$ increases.

177 retained sample $i \in \mathcal{S}$, it recomputes the leave-one-out median $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the induced
 178 distance reduction $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. The k samples with the largest δ_i — i.e.,
 179 those contributing most to the deviation from the reference — are removed and labeled OOD. The
 180 procedure iterates until either the largest attainable reduction falls below the tolerance ϵ or a maximum
 181 iteration count is reached. This stopping rule is a direct consequence of the monotonic trend in Figure
 182 1: as OOD samples are progressively removed, d_t decreases toward a small residual value, at which
 183 point further removals would begin to discard InD samples and degrade detection; halting when the
 184 inter-iteration change in d_t drops below ϵ prevents this over-filtering.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$
 (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for** each $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\hat{\mathcal{S}}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

185 3.2 Training the OOD Detector with Candidate Outliers

186 Given the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ returned by the filtering stage, we estimate the OOD detector g_θ
 187 so as to separate InD from candidate-outlier data, following the training protocol of Du et al. (2024a).
 188 The objective enforces sign separation at the threshold 0: it penalizes non-positive outputs on labeled
 189 InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and positive outputs on candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. Formally, the empirical
 190 risk is

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \tag{5}$$

191 Because the 0/1 losses in (5) are non-differentiable, we replace the indicators with a sigmoid-based
 192 surrogate, yielding a differentiable objective that is amenable to gradient-based optimization while
 193 preserving the sign-separation semantics of the original risk. The overall training loss additionally
 194 retains the InD risk of Equation 2, which regularizes g_θ toward preserving InD predictive accuracy.
 195 Optimizing this joint objective improves the generalization of g_θ and yields robust detection of
 196 samples drawn from $\mathbb{P}_{\text{out}}$, as our experiments confirm.

197 4 Theoretical Analysis

198 We now state the theoretical guarantees for the filtering stage of Medix. The two theorems below
 199 furnish non-asymptotic, high-probability upper bounds on the misclassification rates for InD and
 200 OOD points, respectively, and jointly certify the two-sided robustness of EWM filtering. Full proofs
 201 are deferred to Appendix C.

202 **Theorem 4.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points*

in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

203

204 Theorem 4.1 decomposes the inlier misclassification rate into two additive contributions: a *concentration term* that captures finite-sample fluctuations of the InD gradients and vanishes at rate $O(1/\sqrt{m_{\text{in}}})$,
 205 and a *contamination term* that quantifies the bias induced on the EWM by OOD gradients. Since
 206 the contamination term $\pi/[2(1-\pi)]$ stays finite precisely when $\pi < 0.5$, the bound formalizes the
 207 breakdown behavior of median-based filtering: it tolerates any OOD proportion strictly below one
 208 half. The complementary direction — *what fraction of true outliers escapes detection and is retained*
 209 *as InD?* — is addressed by the following theorem, which bounds the OOD misclassification rate.
 210

Theorem 4.2 (Outlier Misclassification Bound). Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1-\pi}{2\pi}}_{\text{Contamination term}}.$$

211 Theorem 4.2 is the dual of the preceding result, bounding the fraction of OOD points erroneously
 212 retained as InD. Taken together, Theorems 4.1 and 4.2 constitute a two-sided robustness certificate
 213 for Medix filtering. The two misclassification rates are governed by the interplay of the following
 214 three effects.

215 **Contamination Effect.** In Theorem 4.1, the EWM criterion remains stable under any OOD fraction
 216 $\pi < 0.5$; this is encoded by the term $\pi/[2(1-\pi)]$, which is monotonically increasing in π and
 217 measures the bias that OOD gradients impart to the median. Theorem 4.2 carries the symmetric term
 218 $\frac{1-\pi}{2\pi}$, which grows as $\pi \rightarrow 0$ and quantifies the difficulty of isolating OOD points when they are
 219 underrepresented.

220 **Concentration Effect.** Both bounds invoke the sub-Gaussianity of the gradient coordinates. For
 221 InD data, sub-Gaussian tails concentrate most gradients near $\bar{\nabla}_{\text{in}}$, stabilizing the median in the
 222 finite-sample regime; for OOD data, concentration about μ_{out} upper-bounds the probability that a
 223 misaligned OOD gradient evades the filter. Both concentration terms decay as $1/\sqrt{m}$ in the per-class
 224 sample size m .

225 **Separation Effect.** Present only in Theorem 4.2, this effect specifies the minimal separation
 226 between the OOD and InD mean gradients required for reliable rejection. The exponential factor
 227 $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ makes the trade-off explicit: larger mean separation Δ drives this factor to
 228 zero, exponentially suppressing the probability that OOD gradients are misretained.

Theorems 4.1 and 4.2 jointly certify that, with high probability, EWM filtering separates the InD and OOD components of an unlabeled mixture. The InD-to-outlier error is controlled by contamination and concentration terms; the OOD-to-inlier error is controlled by an exponential separation term, a concentration term, and a reverse contamination term. Under the stated (mild) assumptions, both error types are thus provably small.

229

230 **Remark 4.3.** *Sub-Gaussian Assumption for Gradient Coordinates.* The coordinate-wise sub-Gaussian
 231 assumption on InD gradients underpins the concentration terms in both theorems and is what makes
 232 the median estimator statistically stable. We provide two empirical checks of this assumption.

First, Figure 4a shows that the marginal histogram of InD gradient values is unimodal and tightly concentrated about its mean, exhibiting the light (exponentially decaying) tails that characterize sub-Gaussian variables and under which extreme gradient values occur with vanishing probability. Second, the Q-Q plot in Figure 4b (Wilk and Gnanadesikan, 1968) compares the empirical gradient quantiles against the theoretical Gaussian quantiles; the near-linear alignment along the 45° reference line indicates approximate Gaussianity, and since the Gaussian family is a subclass of the sub-Gaussian family, this is sufficient evidence for the assumption. *For completeness, we also derive a relaxed variant of these bounds that **dispenses with sub-Gaussianity** and requires only bounded second moments; it is stated as Theorem C.3 (Appendix C.3). The convergence rates deteriorate under this weaker assumption, but the qualitative robustness guarantee of Medix is preserved.*

5 Experiments

This section reports an empirical evaluation of Medix across a range of InD–OOD dataset pairs, benchmarked against 20 widely used baselines. All experiments are executed on NVIDIA A100-SXM4-80GB GPUs. Code sufficient to reproduce the reported results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. We adopt the experimental protocol of Katz-Samuels et al. (2022a), which introduced the task of learning OOD detectors from wild data, ensuring comparability with prior work. The InD marginals ($\mathbb{P}_{\text{in}}$) are CIFAR-10 and CIFAR-100. The OOD marginals ($\mathbb{P}_{\text{out}}$) are a suite of natural-image datasets: PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), LSUN-RESIZE, and LSUN-C (Yu et al., 2015). The wild distribution ($\mathbb{P}_{\text{wild}}$) is synthesized as a mixture of a subset of $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{out}}$ at the default contamination ratio $\pi = 0.5$. Concretely, with PLACES365 as the OOD component, the wild set combines CIFAR with PLACES365 and evaluation is performed on PLACES365; this construction is repeated for every OOD dataset and every baseline. Each InD CIFAR dataset is partitioned into two disjoint halves: the first 25,000 images fit the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$, and the remaining images form the wild mixture $\mathcal{S}_{\text{wild}}$. Gradients are computed with respect to the penultimate-layer weights, which prior work has shown to be especially informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard metrics: (1) the false positive rate on OOD samples at a fixed 95% true positive rate on InD samples (FPR95 $\downarrow$); (2) the area under the receiver operating characteristic curve (AUROC $\uparrow$); and (3) the InD classification accuracy (InD Acc $\uparrow$). Arrows indicate the direction of improvement.

Baselines. The baselines are partitioned by their training supervision into two groups: those trained on InD data alone and those trained on InD and wild data jointly. The first group (trained only on $\mathbb{P}_{\text{in}}$) comprises established post-hoc and training-time detectors: Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis distance (Lee et al., 2018a), Energy score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all built on a softmax cross-entropy backbone; we additionally include the contrastively trained detectors CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). The second group (trained on both $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$) comprises Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularized learning (Liu et al., 2020b), which regularize training by driving down confidence or up energy on outlier data; WOODS (Katz-Samuels et al., 2022a), which introduced unlabeled wild data and exploits it via constrained optimization; and the more recent CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024).

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ is a Wide ResNet (Zagoruyko, 2016) of depth 40 and width factor 2. It is optimized by stochastic gradient descent with momentum 0.9, weight decay 5×10^{-4} , and initial learning rate 0.1, over 100 epochs under a cosine learning-rate schedule, with batch size 128 and dropout rate 0.3. The Medix hyperparameters ϵ and k are tuned over the grids $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$ respectively, chosen with respect to

dataset size to maximize OOD performance. The OOD detector g_θ is initialized from the pre-trained $\phi_{\mathcal{S}_{\text{in}}}$ and augmented with a linear binary-classification head on the penultimate-layer features. It is fine-tuned for 100 epochs at learning rate 0.001 by minimizing the surrogate of Equation (5), in which the binary loss is combined with the InD classification loss under a relative weight of 10 on the binary term; all remaining optimization settings match those of $\phi_{\mathcal{S}_{\text{in}}}$.

Table 1: OOD detection performance of Medix against baselines with CIFAR-10 as the InD marginal $\mathbb{P}_{\text{in}}$. Entries are means over five independent runs; per-column best values are set in **bold**. FPR95 is reported in percent (lower is better) and AUROC in percent (higher is better).

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{\text{in}}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wid}}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance of Medix against baselines with CIFAR-100 as the InD marginal $\mathbb{P}_{\text{in}}$. Entries are means over five independent runs; per-column best values are set in **bold**. FPR95 is reported in percent (lower is better) and AUROC in percent (higher is better).

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using $\mathbb{P}_{\text{in}}$ only												
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

5.3 Results

The primary results on CIFAR-100 are reported in Table 2, where Medix attains the best value in every column. Two observations follow. (1) Detectors trained on both InD and wild data dominate those trained on InD data alone: relative to KNN+, Medix lowers FPR95 by 52.31 percentage points on PLACES365 and by 38.24 percentage points on TEXTURES, confirming the regularizing value of in-the-wild data. (2) Among wild-data methods ($\mathbb{P}_{\text{wild}}$), Medix remains strictly superior: it achieves an average FPR95 of 5.42%, a 1.32-percentage-point reduction over WOODS, while retaining a competitive InD accuracy of 73.33%. The small accuracy gap is attributable to the training-set sizes: Medix is fit on 25,000 labeled InD samples, whereas the InD-only baselines use the full 50,000-sample CIFAR-100 training set. The CIFAR-10 results in Table 1 exhibit the same ordering, with Medix best on every column; in FPR95 it improves over WOODS by 7.21 percentage points on PLACES365 and 5.25 percentage points on TEXTURES.

A representative visual example of Medix. To visualize the behavior of the outlier-extraction stage (Algorithm 1) on $\mathcal{S}_{\text{wild}}$, we construct a controlled two-dimensional synthetic instance in which ground-truth membership is known. The InD component is a three-class Gaussian mixture with class means $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$ and common isotropic covariance $0.25 \cdot I$; the OOD component is a single Gaussian $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$, well separated from the InD means. As shown in Figure 2, the extracted set aligns closely with the ground-truth OOD points: Medix recovers 87.5% of the true OOD samples as outliers, corresponding to the 12.5% inlier misclassification rate reported above.

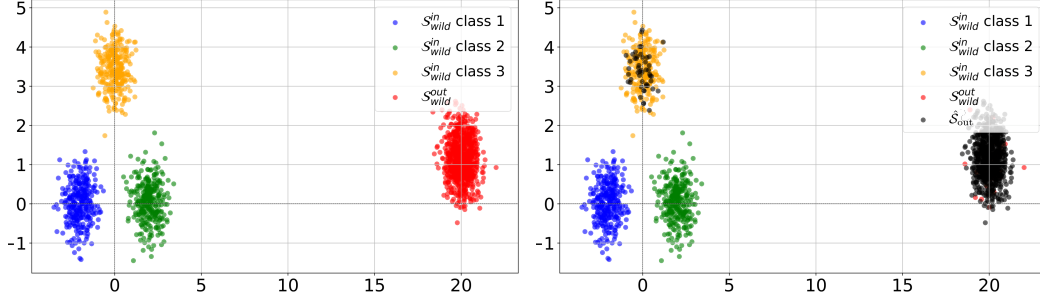


Figure 2: Medix applied to a two-dimensional synthetic wild mixture. (a) Ground-truth configuration of the InD component $\mathcal{S}_{\text{wild}}^{\text{in}}$ and the OOD component $\mathcal{S}_{\text{wild}}^{\text{out}}$, with inliers drawn from a mixture of three multivariate Gaussians. (b) Outlier set $\hat{\mathcal{S}}_{\text{out}}$ returned by Medix (black); the fraction of $\hat{\mathcal{S}}_{\text{out}}$ that is in fact InD (i.e., the inlier misclassification rate) is only 12.5%.

307 **Additional studies and insights.** Owing to space constraints, further experiments and analyses are
 308 deferred to Appendix A: (1) ablations on hyperparameter selection and sensitivity (Appendix A.2),
 309 establishing robustness of Medix to the choice of ϵ and k ; (2) a comparison of the element-wise
 310 median against the geometric median (Appendix A.1), showing that the EWM is more sensitive to
 311 distributional shift and hence a more effective filtering statistic; (3) an expanded comparison against
 312 detectors trained with contrastive objectives (Appendix A.3), on which Medix retains a substantial
 313 margin; (4) an evaluation on large-scale data under the unseen-OOD regime $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix
 314 A.4), where Medix again dominates the baselines; (5) a profiling of the compute and memory cost of
 315 Medix (Appendix A.6); (6) an analysis of the effect of pseudo-label quality (Appendix A.5), showing
 316 resilience to noisy or low-confidence pseudo-labels; and (7) a comparison with semi-supervised
 317 open-set recognition methods (Appendix A.7).

318 6 Related Work

319 In recent years, there has been a growing interest in OOD detection (Fort et al., 2021; Yang et al.,
 320 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023;
 321 Djurisic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al.,
 322 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024;
 323 Abbas et al., 2025). One family of detectors assigns each input a scalar OOD score derived from the
 324 trained model. This family includes distance-based scores (Lee et al., 2018b; Tack et al., 2020b; Ren
 325 et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al.,
 326 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang
 327 et al., 2021; Wu et al., 2023), confidence-based scores (Bendale and Boult, 2016; Hendrycks and
 328 Gimpel, 2017; Liang et al., 2018), and Bayesian uncertainty estimates (Gal and Ghahramani, 2016;
 329 Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020;
 330 Kristiadi et al., 2020).

331 A second family instead modifies the training objective through regularization (Malinin and Gales,
 332 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020;
 333 Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b,
 334 2023; Wang et al., 2023a), for example by penalizing high confidence (Lee et al., 2017; Hendrycks
 335 et al., 2019) or low energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD inputs.
 336 Most such methods presuppose access to a clean auxiliary OOD dataset.

337 A third line (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relaxes the labeled-only
 338 assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025),
 339 but without an explicit outlier-identification mechanism. Du et al. (2024a,b) do introduce explicit
 340 outlier filtering, yet their thresholding rule is fundamentally distinct from our median-centric criterion.
 341 A further methodological distinction is the mixing assumption: Katz-Samuels et al. (2022a); Du
 342 et al. (2024a) assume batch-level mixing, in which each mini-batch contains a fixed InD/OOD ratio,
 343 whereas for large outsourced corpora the InD and OOD samples are interleaved randomly across the
 344 entire dataset. Medix operates directly under this dataset-level mixing and imposes no batch-level

structure. Finally, a body of work on positive-unlabeled learning trains classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022); unlike Medix, these methods only discriminate $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and do not jointly train a dedicated OOD detector, nor do they provide the provably low error rates that our median-centric analysis establishes.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection from unlabeled in-the-wild data. By exploiting the bounded influence of the coordinate-wise median, Medix extracts outliers from the unlabeled mixture and uses them, together with labeled InD data, to fit a reliable OOD detector. On the theoretical side, we proved a finite-sample bound on the inlier misclassification rate, decomposing the error into sub-Gaussian concentration and contamination terms and establishing that the guarantee holds for any OOD contamination ratio $\pi < 0.5$; we complemented this with a dual bound on the OOD misclassification rate under a mean-gradient separation condition. Empirically, Medix outperformed all 20 baselines across the benchmarks: it reduced the average FPR95 by 40.98 percentage points relative to KNN+, surpassed state-of-the-art wild-data methods such as WOODS and DRL, and attained an outlier-extraction error rate as low as 12.5%, in quantitative agreement with the theory.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

We next justify the choice of the element-wise median over the geometric median (Acharya et al., 2024) as the robust aggregation statistic in Medix. To this end, we conduct a controlled experiment with CIFAR-10 as InD and SVHN as OOD. We initialize $\mathcal{S}_{\text{wild}}$ with 5,000 CIFAR-10 samples held disjoint from the data used to fit $\phi_{\mathcal{S}_{\text{in}}}$ and to compute $\bar{\nabla}_{\text{in}}$, then inject SVHN OOD samples incrementally, tracking both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the wild-set gradient median and the proportion of OOD samples flagged. As Figure 3 shows, at any fixed OOD count the element-wise median flags a strictly larger proportion of OOD samples than the geometric median. The element-wise statistic is therefore more sensitive to the distributional shift induced by OOD contamination, which makes it the more effective and reliable filtering statistic in our setting.

A.2 Hyperparameter Selection and Sensitivity Analysis

We assess the sensitivity of Medix to the hyperparameters ϵ and k through an ablation on CIFAR-100 (InD) with SVHN (OOD). As Table 3 shows, the detection performance is stable across the tested grid. At $\epsilon = 0.005$, $k = 20000$, Medix attains an FPR95 of 0.16%. Perturbing either hyperparameter — reducing ϵ by an order of magnitude to 0.0005, or halving k to 10000 — degrades FPR95 only mildly, to 0.20% and 0.68% respectively, i.e., a graceful rather than abrupt degradation. Thus, although the optimum is attained at a specific setting, Medix remains competitive with and superior to the baselines across the hyperparameter range. This insensitivity is practically important, since exhaustive tuning is often infeasible in deployment.

A.3 Comprehensive comparison with recent competitive methods

We extend the comparison to strong detectors trained under contrastive objectives — SSD+ (Schwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023) — which are included despite their differing training paradigm to broaden the evaluation. We further add the recent CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). This experiment uses CIFAR-100 as InD and a ResNet-34 backbone¹, following the training setup of Ming et al. (2023); Zhang et al. (2024): stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs, initial learning rate 0.5 under cosine scheduling, and batch size 512. We

¹ResNet-34 is the standard architecture in these works; we adopt it to ensure a consistent and fair comparison.

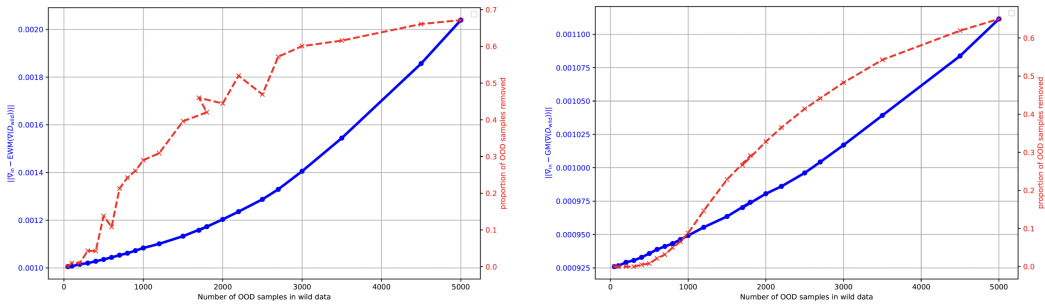


Figure 3: Fraction of injected OOD samples flagged as outliers by the element-wise median (EWM) versus the geometric median (GM), as a function of the OOD count in $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD).

Table 4: OOD detection performance of Medix against recent competitive baselines with CIFAR-100 as the InD marginal $\mathbb{P}_{\text{in}}$ (ResNet-34 backbone). Entries are means over five independent runs; per-column best values are set in **bold**. FPR95 and AUROC are in percent.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

reuse the checkpoints released by Ming et al. (2023)². As reported in Table 4, Medix outperforms every baseline by a substantial margin — including the contrastively trained detectors — attaining an average FPR95 of 8.68% and an average AUROC of 97.76%. In addition, and unlike most of these methods, Medix carries a provable error-rate guarantee.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We now assess Medix on large-scale wild OOD data under the unseen-OOD regime, which stresses the method along two axes: (1) the wild OOD component ($\mathbb{P}_{\text{out}}$) is large-scale, increasing both distributional complexity and computational cost; and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately mismatched from the training-time wild OOD distribution, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, so that generalization to previously unseen OOD sources is required — a more faithful model of deployment. The labeled InD data is CIFAR-100, the unlabeled wild OOD data is the 300K Random Images dataset Hendrycks et al. (2019), and the test OOD data is SVHN. The distributional gap between the wild OOD source (300K Random Images) and the test OOD source (SVHN) is what makes this setup demanding. We compare against the baselines that likewise train on wild data. As Table 5 reports, Medix dominates these baselines, attaining both a markedly lower FPR (41.29±1.2) and a higher AUROC (87.25±0.9), confirming that its InD/OOD separation transfers to unseen OOD distributions.

Table 5: OOD detection performance on large-scale unseen OOD data (CIFAR-100 InD; 300K Random Images as wild OOD; SVHN as test OOD). Values are mean $\pm$ standard deviation; best per column in **bold**. FPR and AUROC in percent.

Method	FPR ↓	AUROC ↑
OE	68.80 $\pm$ 2.8	82.89 $\pm$ 1.1
Energy (w/ OE)	69.81 $\pm$ 2.4	85.59 $\pm$ 1.0
WOODS	69.41 $\pm$ 2.7	86.76 $\pm$ 0.8
Medix (ours)	41.29 $\pm$ 1.2	87.25 $\pm$ 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

We next examine whether the quality of the pseudo-labels $\hat{y}_{\tilde{x}_i}$, as proxied by softmax confidence, affects filtering performance, and whether pre-filtering low-confidence pseudo-labels improves robustness. We test this on CIFAR-10 (InD) with LSUN-Resize (OOD) by computing the softmax probability of each pseudo-labeled wild sample and discarding those below a confidence threshold of 0.6; this removes 15.98% of the training samples.

The two configurations are statistically indistinguishable: both attain an FPR95 of 0.01%, and confidence pre-filtering raises AUROC only marginally, from 99.98% to 99.99%. Discarding low-

²<https://github.com>

confidence pseudo-labels therefore has negligible effect on detection, indicating that Medix is resilient to noisy or low-confidence pseudo-labels and that additional confidence-based pre-filtering is unlikely to yield meaningful gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We now quantify the compute and memory footprint of the Medix filtering stage at scale. We profile the stage on an NVIDIA A100-SXM4-80GB GPU with approximately 15,000 unlabeled samples per InD–OOD pair, using CIFAR-10 and CIFAR-100 as InD and LSUN-Resize as OOD. As Table 6 reports, peak GPU memory stays modest and the filtering stage terminates within a tractable wall-clock budget even at this sample count. The filtering stage is thus computationally feasible and incurs no excessive memory overhead on large datasets.

Table 6: Compute and memory profile of the Medix filtering stage on an NVIDIA A100 80GB GPU, per InD–OOD pair ($\approx 15,000$ unlabeled samples).

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

700

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix differs from recent semi-supervised open-set recognition (SSL) methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) along three axes: objective, treatment of outliers, and mechanism. First, the objectives differ. In OpenMatch Saito et al. (2021), the target is robust training of a standard classifier under outlier contamination in SSL; in our setting, the target is to detect OOD samples in unlabeled wild data and to fit a dedicated OOD detector. Thus SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) optimize classification accuracy in the presence of outliers, whereas Medix optimizes open-world detection without any labeled OOD data.

Second, the treatment of outliers differs. SSL methods regard outliers as nuisance noise to be suppressed during training, so as to improve classification. Medix instead treats the same outliers as signal: it explicitly identifies them and reuses them as negative supervision for the detector.

Third, the mechanisms differ. OpenMatch is built on consistency regularization, imposing soft constraints that enforce invariance of the classifier’s outputs across stochastic input transformations and thereby smoothing the decision boundary. Medix is built on median-based filtering, exploiting the statistical robustness of the median — its insensitivity to extreme values — to flag OOD samples, with error bounds derived from the contamination effect (the OOD proportion) and the concentration effect (sub-Gaussianity of InD gradients) that certify a low detection error.

B Broader Impacts and Limitations

Reliable OOD detection is a prerequisite for the safe deployment of machine learning systems under distribution shift. The median-centric framework introduced here improves the handling of OOD inputs and thereby model safety, with downstream applications in safety-critical domains such as healthcare, autonomous systems, and finance; strengthening OOD detection directly mitigates a principal failure mode of deployed models and supports greater trust in their adoption. Promising extensions include coupling the filtering stage with generative models for outlier synthesis and adapting it to non-stationary settings in which the OOD distribution drifts over time.

A principal limitation of Medix is its reliance on an unlabeled mixture of InD and OOD data that may, in practice, be noisy, corrupted, or non-i.i.d. This assumption, however, is standard in robust learning theory and reflects the irreducible uncertainty of deployment. To keep the assumptions explicit and checkable, we formalize them in the main theorems and state, in Remark 4.3, the precise geometric separation condition required for reliable filtering. We further relax the analysis in Theorem C.3, which dispenses with sub-Gaussian tails and retains the guarantee under a bounded-moment condition

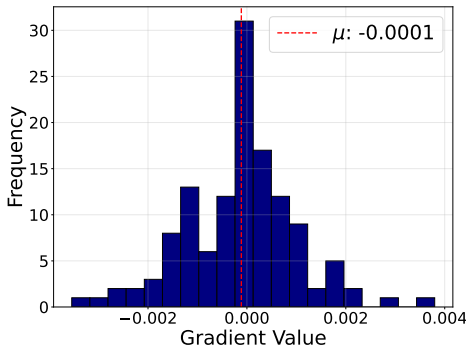
733 alone. These results show that the simplifying assumptions of Medix are both interpretable and
734 relaxable, and thus constitute a principled basis for extending OOD detection to more realistic noise
735 regimes.

736 **C Proof of Theorem**

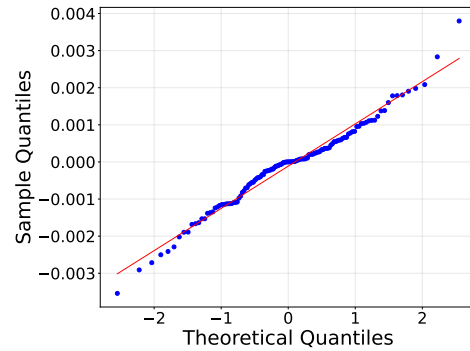
737 For reference throughout the theoretical and empirical sections, Table 7 collects the notation used
 738 in this paper, covering gradient-related quantities, sample-size parameters, filtering terms, and the
 739 parameters appearing in the concentration bounds. Unless stated otherwise, boldface denotes vectors,
 740 $\mathcal{X}$ denotes the input space, and all gradients are taken with respect to a fixed pretrained model f_ϕ .

Table 7: Summary of notation. Bold symbols denote vectors and $\mathcal{X}$ denotes the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Empirical evidence for coordinate-wise sub-Gaussianity of InD gradients. (Left) Histogram of InD gradient values (CIFAR-100), concentrated about the mean with light tails, as expected under sub-Gaussianity. (Right) Q-Q plot of the empirical InD-gradient quantiles against theoretical Gaussian quantiles; the near-linear alignment supports the sub-Gaussian assumption.

741 C.1 Bound on the Inlier Misclassification Rate

742 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 743 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and*
 744 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 745 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

746 *Proof.* The argument proceeds in two steps. **Step 1** establishes a high-probability concentration
 747 bound placing the gradient of each InD sample within a fixed radius of the mean $\bar{\nabla}_{\text{in}}$; an InD sample
 748 satisfying this bound is termed a *good inlier*. **Step 2** applies a “swapping” argument to the EWM
 749 objective, showing that excluding more than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good inliers* — where η is
 750 the fraction of InD samples falling outside the high-probability region — would strictly improve the
 751 objective and hence contradict optimality of the minimizer. Combining the two steps yields the stated
 752 upper bound on the inlier misclassification rate ERR_{in} .

753 **Step 1: Concentration of InD Gradients.** Let $\nabla\ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 754 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 755 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla\ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

756 Define

$$\mathbf{v} := \nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

757 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

758 so if $\|\mathbf{v}\|_2 > \epsilon\sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

759 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

760 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 761 $\mathbf{x}$ to be *bad* if

$$\|\nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}. \quad (8)$$

762 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

763 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 764 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

765 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

766 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 767 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla\ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

768 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

769 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

770 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
771 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

772 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log\left(\frac{1}{\delta}\right)}{2}}, \quad (17)$$

773 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

774 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

775 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

776 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
777 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
778 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
779 called *good inliers*.

780 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
781 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

782 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
783 thereby contradicting the optimality of $\mathcal{S}^*$.

784 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 785 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

786 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

787 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

788 Define the swap size as

$$m'_\varepsilon := \min \{ m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (26)$$

789 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 790 available for swapping in.

791 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 792 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

793 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

794 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

795 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

796 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

797 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 798 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

799 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

800 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

801 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or increase it, so the median distance from μ_j does not increase and possibly decreases.

Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes the proof.

$$\Rightarrow \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

□

C.2 Bound on the Outlier Retention Rate

We now derive a non-asymptotic upper bound on the fraction of OOD points retained in the subset returned by the MEDIX filter. The argument parallels the inlier misclassification bound (Appendix C.1), except that inlier concentration is replaced by a separation assumption on the OOD and InD gradient means as the driving condition.

Theorem C.2 (Outlier Retention Bound under Vector Separation). *Assume that for every OOD point $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:*

(I) **Sub-Gaussianity.** *Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .*

(II) **Vector Mean Separation.** *The mean OOD gradient vector μ_{out} satisfies*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

827 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

828 **Proof. Step 1: Concentration of OOD Gradients.**

829 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
830 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

831 has probability at most $p_{\text{out}}(\varepsilon)$.

832 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

833 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

834 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

835 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

836 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

837 which holds with probability at least $1 - \delta$.

838 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

839 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_{\varepsilon}^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

840 Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_{\varepsilon} := \left\{g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}. \quad (46)$$

841 By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of
842 entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

843 Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal
 844 number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$
 845 are ambiguous, and the total number of InD points is m_{in} .

846 Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$,
 847 implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

848 Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

849 This strict improvement contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, so the assumption cannot hold. Conse-
 850 quently, with probability at least $1 - \delta$, the outlier retention rate satisfies

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

851

□

852 C.3 Theorem C.3 without Sub-Gaussianity

853 **Theorem C.3** (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients*
 854 *of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment*
 855 *bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

856 Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and
 857 confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of
 858 the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

859 **Comparison with the sub-Gaussian bound.** The two bounds differ only in their dimension-
 860 dependent decay term. Under sub-Gaussianity, this term is

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right),$$

861 which decays exponentially in the gradient dimension d and is therefore extremely tight for large d .
 862 Under the weaker finite-fourth-moment assumption, it is replaced by

$$\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2},$$

863 which decays only as $O(1/d)$ and thus yields a looser bound at large d . The trade-off is one of tail
 864 assumptions: the sub-Gaussian bound requires exponentially light tails, whereas the moment-based
 865 bound admits heavier tails. The finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}$$

866 is common to both and decays in m_{in} independently of the tail condition.

867 *Proof.* The proof retains the two-step structure of Theorem 4.1: (1) a concentration bound on the
 868 fraction of “bad” InD samples (those far from ∇_{in}), now derived without sub-Gaussianity, and (2) a
 869 swapping argument bounding the inlier misclassification rate. The only substantive change is in Step
 870 1, where the sub-Gaussian tail bound is replaced by a moment-based bound obtained via Chebyshev’s
 871 inequality under the finite-fourth-moment assumption.

872 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

873 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 874 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

875 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 876 S_{wild} .

877 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 878 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

879 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

880 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

881 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

882 Using Chebyshev’s inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

883 We need $\mathbb{P}(\|v\|_2 > \epsilon \sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

884 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

885 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}\left(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

886 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
887 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

888 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
889 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

890 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

891 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

892 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

893 This η represents the upper bound on the fraction of InD samples whose gradients deviate from ∇_{in}
894 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

895 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
896 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1 - \pi)}, \quad (68)$$

897 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 898 thereby contradicting the optimality of $\mathcal{S}^*$.

899 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 900 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (69)$$

901 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

902 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

903 Define the swap size as

$$m'_\varepsilon := \min \{ m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

904 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 905 available for swapping in.

906 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 907 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

908 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

909 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

910 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

911 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

912 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 913 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

914 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

915 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

916 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

917 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

918 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

919 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
920 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

921 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
922 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
923 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

924 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

925 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
926 increase it, so the median distance from μ_j does not increase and possibly decreases.

927 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

928 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

929 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
930 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

931

□